

What is a web service ?

A solution to distributed computing.

Integral part of SOA (service oriented architecture)

service = Business functionality to be exported to remote clients via standard protocols(HTTP/s)

server -- service provider

client -- service accessor

Why -- To export the Business logic (functional logic --banking, customer service, payment gateway , stock exchange server BSE , NSE ...) to remote clients over standard set of protocols.

Its equivalent to Java RMI (remote method invocation)

In Java RMI -- java client object can directly invoke the remote method (hosted on the remote host) & get the process results. (i.e it gives you location transparency)

BUT Java RMI ---is 100% java solution.

There is no interoperability in that(i.e its a technology specific solution)

How to arrive at technology independent solution ?

CORBA --- Common object request broker architecture

tough to set up. (IDL ---i/f def language)

Better alternative --- web services

In 2002, the World Wide Web consortium(W3C) had released the definition of WSDL(web service definition language) and SOAP web services. This formed the standard of how web services are implemented.

Earlier (J2EE 1.4) -- JAX-RPC

Java API for XML based remote procedure calls

Today replaced by JAX-WS

In 2004, the web consortium also released the definition of an additional standard called RESTful. Over the past couple of years, this standard has become quite popular. And is being used by many of the popular websites around the world which include Facebook and Twitter.

Corresponding J2EE specification JAX RS

JAX WS -- Java API for XML based web services -- Based upon

Protocol --SOAP -- simple object access protocol (runs over HTTP)

Has additional header & message format .

+ Have to set up Naming service (UDDI --Universal Description, Discovery, and Integration)

+ Have to set up WSDL (web service def. language)-- xml based web service def lang.

--supports only XML as data exchange format.

Too much to set up & eats up larger bandwidth !!

So a simple solution is JAX RS -- Java API for RESTful web service

JAX RS --- is a part of J2EE specifications

Known Vendors --Apache , JBoss

& products --RESTEasy,Apache CXF

BUT it's still difficult to set up.

So spring , being integration master , comes to the rescue....